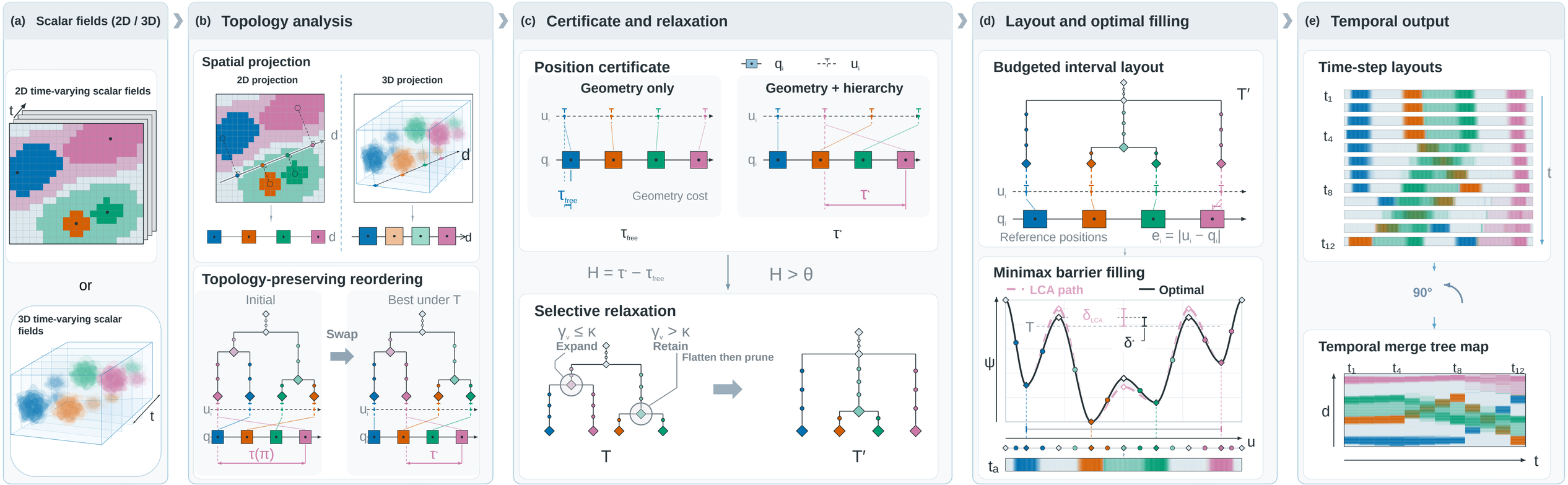

(a) Scalar fields (2D / 3D)
2D time-varying scalar fields
t
or
3D time-varying scalar fields
t
(b) Topology analysis
Spatial projection
2D projection
3D projection
d
d
d
d
Topology-preserving reordering
Initial
Best under T
Swap
u_i
q_i
$\tau(\pi)$
u_i
q_i
τ^*
(c) Certificate and relaxation
Position certificate
q_i
u_i
Geometry only
Geometry + hierarchy
u_i
q_i
τ_{free}
Geometry cost
u_i
q_i
τ^*
τ_{free}
τ^*
$H = \tau^* - \tau_{free}$
$H > \theta$
Selective relaxation
$\gamma_v \le K$
Expand
$\gamma_v > K$
Retain
Flatten then prune
T
T′
(d) Layout and optimal filling
Budgeted interval layout
T′
u_i
q_i
Reference positions
$e_i = |u_i - q_i|$
Minimax barrier filling
LCA path
Optimal
δ_{LCA}
T
δ^*
ψ
u
t_a
(e) Temporal output
Time-step layouts
t_1
t_4
t_8
t_{12}
t
90°
Temporal merge tree map
t_1
t_4
t_8
t_{12}
d
t